REPORT TECNICO DI PROGETTO: Architettura di Rete Campus ad Alta Affidabilità (High-Availability Collapsed Core)

1. Panoramica dell'Architettura

Il progetto implementa un'infrastruttura di rete aziendale suddivisa in due sedi geografiche con un modello di distribuzione Collapsed Core a ridondanza elevata e servizi DHCP integrati sugli switch core per la gestione dinamica degli indirizzi IP:

Motivazione del Modello Collapsed Core (Architettura a 2 Livelli)

Per l'architettura di questo campus distribuito su due sedi, è stata preferita la topologia Collapsed Core rispetto al classico modello a tre livelli (Core, Distribution, Access). Questa scelta ingegneristica offre rilevanti vantaggi operativi e prestazionali:

- **Riduzione della latenza**: La fusione del livello di Core e di Distribuzione in un unico layer multilivello riduce il numero di hop di routing per il traffico inter-VLAN e inter-sede, migliorando la velocità di risposta delle applicazioni.
- **Ottimizzazione dei costi e della gestione**: Abbatte significativamente i costi di acquisizione hardware, consumo energetico e manutenzione, semplificando la topologia di rete senza compromettere le funzionalità.
- **Eliminazione dei Single Point of Failure (SPOF)**: La ridondanza attiva-passiva a livello di gateway (HSRP) unita alle doppie dorsali in fibra ottica tra le sedi garantisce continuità di servizio assoluta anche in caso di guasto hardware di un intero switch Core o di un cavo di dorsale.

- **Sede 1 (Edificio principale)**: Ospita il nocciolo della rete (Core 1 e Core 2), il Firewall Cisco ASA 5506-X di perimetro, il Router Edge (ISR4321) e la DMZ (con il server DVWA). Gestisce 6 VLAN dipartimentali (VLAN 10-60).
- **Sede 2 (Edificio secondario)**: Gestita da una coppia di switch core gemelli (Core 1b e Core 2b), distribuiti su 3 piani con le relative VLAN di piano (VLAN 70-90). Il traffico della Sede 2 viene instradato verso la Sede 1 tramite un collegamento in fibra ottica ridondato.
- **Servizi DHCP Integrati**: I Multilayer Switch di ciascuna sede fungono da server DHCP locali per le rispettive VLAN gestite. Vengono definiti i pool di indirizzi con esclusione dei primi IP riservati alle interfacce SVI e agli indirizzi virtuali HSRP.

Sistemi e Tecnologie Utilizzate

- **Multilayer Switch (Cisco L3)**: Agiscono come centro nevralgico della rete, combinando la commutazione Layer 2 ad altissima velocità con l'instradamento Layer 3 inter-VLAN eseguito direttamente a livello hardware (ASIC). L'integrazione dei server DHCP su di essi azzera la latenza nell'assegnazione degli indirizzi IP.

- **HSRP (Hot Standby Router Protocol)**: Fornisce ridondanza trasparente per i gateway predefiniti delle VLAN. In caso di malfunzionamento dello switch primario, lo switch secondario subentra in pochi millisecondi senza alcuna interruzione per gli utenti.
- **Rapid-PVST+ (Per-VLAN Spanning Tree Plus)**: Previene la formazione di loop Layer 2 sui collegamenti trunk ridondati e assicura tempi di convergenza della topologia estremamente rapidi in caso di guasto di un link.
- **Firewall Cisco ASA 5506-X**: Presidia la sicurezza perimetrale gestendo tre distinte zone di fiducia: Inside (livello 100, rete interna protetta), DMZ (livello 50, servizi pubblici/isolati) e Outside (livello 0, rete esterna/Internet). Applica ispezione stateful del traffico e Port Address Translation (PAT) dinamico.
- **Router Edge (ISR4321)**: Termina la connessione verso il provider Internet (WAN/ISP) gestendo il routing puro verso l'esterno e lasciando la sicurezza perimetrale interamente delegata all'ASA.

2. Configurazione e Cablaggio dei Multilayer Switch

Schema dei Collegamenti Fisici

La robustezza della rete si basa su una doppia simmetria di collegamento:

- **Dorsale in Fibra Inter-sede (Principale)**: Core 1 (Sede 1) ↔ Core 1b (Sede 2) collegati sulla porta in fibra GigabitEthernet1/1/1 su entrambi i lati.
- **Dorsale in Fibra Inter-sede (Backup)**: Core 2 (Sede 1) ↔ Core 2b (Sede 2) collegati sulla porta in fibra GigabitEthernet1/1/1 su entrambi i lati.
- **Ring / Trunk Locale Intra-sede**: In ciascuna sede, i due switch core locali sono interconnessi direttamente tramite la porta GigabitEthernet1/1/2 configurata in modalità Trunk, garantendo la continuità del piano di controllo (HSRP) e del traffico locale.
- **Distribuzione di Piano**: Le porte da GigabitEthernet1/0/1 a 1/0/9 gestiscono i collegamenti trunk verso gli switch di accesso dei vari piani.

Configurazioni CLI dei Multilayer Switch

Core 1 (Sede 1 / Primario)

enable

configure terminal

ip routing

spanning-tree mode rapid-pvst

ip dhcp excluded-address 192.168.10.1 192.168.10.10

ip dhcp excluded-address 192.168.20.1 192.168.20.10

```
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.50.1 192.168.50.10
ip dhcp excluded-address 192.168.60.1 192.168.60.10
ip dhcp pool VLAN10_POOL
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool VLAN20_POOL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
ip dhcp pool VLAN30_POOL
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
ip dhcp pool VLAN40_POOL
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
 dns-server 8.8.8.8
ip dhcp pool VLAN50_POOL
 network 192.168.50.0 255.255.255.0
 default-router 192.168.50.1
```

```
 dns-server 8.8.8.8
ip dhcp pool VLAN60_POOL
 network 192.168.60.0 255.255.255.0
 default-router 192.168.60.1
 dns-server 8.8.8.8
vlan 10,20,30,40,50,60
interface Vlan10
 ip address 192.168.10.2 255.255.255.0
 standby 10 ip 192.168.10.1
 standby 10 priority 110
 standby 10 preempt
 no shutdown
interface Vlan20
 ip address 192.168.20.2 255.255.255.0
 standby 20 ip 192.168.20.1
 standby 20 priority 110
 standby 20 preempt
 no shutdown
interface Vlan30
 ip address 192.168.30.2 255.255.255.0
 standby 30 ip 192.168.30.1
 standby 30 priority 110
 standby 30 preempt
```

```
 no shutdown
interface Vlan40
 ip address 192.168.40.2 255.255.255.0
 standby 40 ip 192.168.40.1
 standby 40 priority 110
 standby 40 preempt
 no shutdown
interface Vlan50
 ip address 192.168.50.2 255.255.255.0
 standby 50 ip 192.168.50.1
 standby 50 priority 110
 standby 50 preempt
 no shutdown
interface Vlan60
 ip address 192.168.60.2 255.255.255.0
 standby 60 ip 192.168.60.1
 standby 60 priority 110
 standby 60 preempt
 no shutdown
spanning-tree vlan 10,20,30,40,50,60 root primary
interface range GigabitEthernet1/0/1 - 9
 switchport mode trunk
 no shutdown
```

interface GigabitEthernet1/1/2

 switchport mode trunk

 no shutdown

interface GigabitEthernet1/0/10

 no switchport

 ip address 192.168.210.2 255.255.255.252

 no shutdown

interface GigabitEthernet1/1/1

 no switchport

 ip address 192.168.200.1 255.255.255.252

 no shutdown

ip route 192.168.70.0 255.255.255.0 192.168.200.2

ip route 192.168.80.0 255.255.255.0 192.168.200.2

ip route 192.168.90.0 255.255.255.0 192.168.200.2

ip route 0.0.0.0 0.0.0.0 192.168.210.1

end

copy running-config startup-config

Core 2 (Sede 1 / Secondario)

enable

configure terminal

ip routing

spanning-tree mode rapid-pvst

ip dhcp excluded-address 192.168.10.1 192.168.10.10

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.50.1 192.168.50.10
ip dhcp excluded-address 192.168.60.1 192.168.60.10
ip dhcp pool VLAN10_POOL
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool VLAN20_POOL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
ip dhcp pool VLAN30_POOL
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
ip dhcp pool VLAN40_POOL
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
 dns-server 8.8.8.8
ip dhcp pool VLAN50_POOL
 network 192.168.50.0 255.255.255.0
```

```
 default-router 192.168.50.1
 dns-server 8.8.8.8
ip dhcp pool VLAN60_POOL
 network 192.168.60.0 255.255.255.0
 default-router 192.168.60.1
 dns-server 8.8.8.8
vlan 10,20,30,40,50,60
interface Vlan10
 ip address 192.168.10.3 255.255.255.0
 standby 10 ip 192.168.10.1
 standby 10 priority 100
 standby 10 preempt
 no shutdown
interface Vlan20
 ip address 192.168.20.3 255.255.255.0
 standby 20 ip 192.168.20.1
 standby 20 priority 100
 standby 20 preempt
 no shutdown
interface Vlan30
 ip address 192.168.30.3 255.255.255.0
 standby 30 ip 192.168.30.1
 standby 30 priority 100
```

```
 standby 30 preempt
 no shutdown
interface Vlan40
 ip address 192.168.40.3 255.255.255.0
 standby 40 ip 192.168.40.1
 standby 40 priority 100
 standby 40 preempt
 no shutdown
interface Vlan50
 ip address 192.168.50.3 255.255.255.0
 standby 50 ip 192.168.50.1
 standby 50 priority 100
 standby 50 preempt
 no shutdown
interface Vlan60
 ip address 192.168.60.3 255.255.255.0
 standby 60 ip 192.168.60.1
 standby 60 priority 100
 standby 60 preempt
 no shutdown
spanning-tree vlan 10,20,30,40,50,60 root secondary
interface range GigabitEthernet1/0/1 - 9
 switchport mode trunk
```

```
 no shutdown
interface GigabitEthernet1/1/2
 switchport mode trunk
 no shutdown
interface GigabitEthernet1/0/10
 no switchport
 ip address 192.168.211.2 255.255.255.252
 no shutdown
interface GigabitEthernet1/1/1
 no switchport
 ip address 192.168.200.5 255.255.255.252
 no shutdown
ip route 192.168.70.0 255.255.255.0 192.168.200.6
ip route 192.168.80.0 255.255.255.0 192.168.200.6
ip route 192.168.90.0 255.255.255.0 192.168.200.6
ip route 0.0.0.0 0.0.0.0 192.168.211.1 10
end
copy running-config startup-config
```

Core 1b (Sede 2 / Primario)

```
enable
configure terminal
ip routing
spanning-tree mode rapid-pvst
ip dhcp excluded-address 192.168.70.1 192.168.70.10
```

```
ip dhcp excluded-address 192.168.80.1 192.168.80.10
ip dhcp excluded-address 192.168.90.1 192.168.90.10
ip dhcp pool VLAN70_POOL
 network 192.168.70.0 255.255.255.0
 default-router 192.168.70.1
 dns-server 8.8.8.8
ip dhcp pool VLAN80_POOL
 network 192.168.80.0 255.255.255.0
 default-router 192.168.80.1
 dns-server 8.8.8.8
ip dhcp pool VLAN90_POOL
 network 192.168.90.0 255.255.255.0
 default-router 192.168.90.1
 dns-server 8.8.8.8
vlan 70,80,90
interface Vlan70
 ip address 192.168.70.2 255.255.255.0
 standby 70 ip 192.168.70.1
 standby 70 priority 110
 standby 70 preempt
 no shutdown
interface Vlan80
 ip address 192.168.80.2 255.255.255.0
```

```
 standby 80 ip 192.168.80.1
 standby 80 priority 110
 standby 80 preempt
 no shutdown
interface Vlan90
 ip address 192.168.90.2 255.255.255.0
 standby 90 ip 192.168.90.1
 standby 90 priority 110
 standby 90 preempt
 no shutdown
spanning-tree vlan 70,80,90 root primary
interface range GigabitEthernet1/0/1 - 9
 switchport mode trunk
 no shutdown
interface GigabitEthernet1/1/2
 switchport mode trunk
 no shutdown
interface GigabitEthernet1/1/1
 no switchport
 ip address 192.168.200.2 255.255.255.252
 no shutdown
ip route 0.0.0.0 0.0.0.0 192.168.200.1
end
```

copy running-config startup-config

Core 2b (Sede 2 / Secondario)

enable

configure terminal

ip routing

spanning-tree mode rapid-pvst

ip dhcp excluded-address 192.168.70.1 192.168.70.10

ip dhcp excluded-address 192.168.80.1 192.168.80.10

ip dhcp excluded-address 192.168.90.1 192.168.90.10

ip dhcp pool VLAN70_POOL

 network 192.168.70.0 255.255.255.0

 default-router 192.168.70.1

 dns-server 8.8.8.8

ip dhcp pool VLAN80_POOL

 network 192.168.80.0 255.255.255.0

 default-router 192.168.80.1

 dns-server 8.8.8.8

ip dhcp pool VLAN90_POOL

 network 192.168.90.0 255.255.255.0

 default-router 192.168.90.1

 dns-server 8.8.8.8

vlan 70,80,90

interface Vlan70

```
 ip address 192.168.70.3 255.255.255.0
 standby 70 ip 192.168.70.1
 standby 70 priority 100
 standby 70 preempt
 no shutdown
interface Vlan80
 ip address 192.168.80.3 255.255.255.0
 standby 80 ip 192.168.80.1
 standby 80 priority 100
 standby 80 preempt
 no shutdown
interface Vlan90
 ip address 192.168.90.3 255.255.255.0
 standby 90 ip 192.168.90.1
 standby 90 priority 100
 standby 90 preempt
 no shutdown
spanning-tree vlan 70,80,90 root secondary
interface range GigabitEthernet1/0/1 - 9
 switchport mode trunk
 no shutdown
interface GigabitEthernet1/1/2
 switchport mode trunk
```

no shutdown

interface GigabitEthernet1/1/1

no switchport

ip address 192.168.200.6 255.255.255.252

no shutdown

ip route 0.0.0.0 0.0.0.0 192.168.200.5

end

copy running-config startup-config

3. Collegamento e Configurazione del Firewall Cisco ASA 5506-X

Topologia delle Interfacce ASA

Interfaccia	Nome Interfaccia	Livello Sicurezza	Subnet / IP	Descrizione / Destinazione
GigabitEthernet1/1	`inside-s1-prim`	100	`192.168.210.1/30`	Core 1 di Sede 1 (Percorso primario)
GigabitEthernet1/2	`inside-s1-sec`	100	`192.168.211.1/30`	Core 2 di Sede 1 (Percorso di ridondanza)
GigabitEthernet1/3	`outside`	0	`192.168.100.2/30`	Connection to Router Edge
GigabitEthernet1/4	`dmz`	50	`192.168.150.1/24`	Zona isolata per server (es. server DVWA)

Configurazione CLI del Firewall ASA

enable

configure terminal

clear configure interface GigabitEthernet1/1

clear configure interface GigabitEthernet1/2

clear configure interface GigabitEthernet1/3

clear configure interface GigabitEthernet1/4

```
clear configure nat

clear configure object

clear configure route

same-security-traffic permit inter-interface

interface GigabitEthernet1/1

 nameif inside-s1-prim

 security-level 100

 ip address 192.168.210.1 255.255.255.252

 no shutdown

interface GigabitEthernet1/2

 nameif inside-s1-sec

 security-level 100

 ip address 192.168.211.1 255.255.255.252

 no shutdown

interface GigabitEthernet1/3

 nameif outside

 security-level 0

 ip address 192.168.100.2 255.255.255.252

 no shutdown

interface GigabitEthernet1/4

 nameif dmz

 security-level 50

 ip address 192.168.150.1 255.255.255.0
```

```
 no shutdown

route outside 0.0.0.0 0.0.0.0 192.168.100.1 1

route inside-s1-prim 192.168.10.0 255.255.255.0 192.168.210.2 1

route inside-s1-prim 192.168.20.0 255.255.255.0 192.168.210.2 1

route inside-s1-prim 192.168.30.0 255.255.255.0 192.168.210.2 1

route inside-s1-prim 192.168.40.0 255.255.255.0 192.168.210.2 1

route inside-s1-prim 192.168.50.0 255.255.255.0 192.168.210.2 1

route inside-s1-prim 192.168.60.0 255.255.255.0 192.168.210.2 1

route inside-s1-prim 192.168.70.0 255.255.255.0 192.168.210.2 1

route inside-s1-prim 192.168.80.0 255.255.255.0 192.168.210.2 1

route inside-s1-prim 192.168.90.0 255.255.255.0 192.168.210.2 1

route inside-s1-sec 192.168.10.0 255.255.255.0 192.168.211.2 10

route inside-s1-sec 192.168.20.0 255.255.255.0 192.168.211.2 10

route inside-s1-sec 192.168.30.0 255.255.255.0 192.168.211.2 10

route inside-s1-sec 192.168.40.0 255.255.255.0 192.168.211.2 10

route inside-s1-sec 192.168.50.0 255.255.255.0 192.168.211.2 10

route inside-s1-sec 192.168.60.0 255.255.255.0 192.168.211.2 10

route inside-s1-sec 192.168.70.0 255.255.255.0 192.168.211.2 10

route inside-s1-sec 192.168.80.0 255.255.255.0 192.168.211.2 10

route inside-s1-sec 192.168.90.0 255.255.255.0 192.168.211.2 10

object network RETE-INTERNA

 subnet 192.168.0.0 255.255.0.0

 nat (inside-s1-prim,outside) dynamic interface
```

nat (inside-s1-sec,outside) dynamic interface

end

write memory

4. Configurazione del Router Edge (ISR4321)

Il router gestisce la connessione verso l'esterno (Cloud/ISP) con rotta di ritorno verso l'ASA:t
enable
configure terminal
hostname RTV-EDGE

ip routing

interface GigabitEthernet0/0/0
description LINK-TO-ASA-OUTSIDE
ip address 192.168.100.1 255.255.255.252
no shutdown

interface GigabitEthernet0/0/1
description LINK-TO-CLOUD-ISP
ip address dhcp
no shutdown

ip route 192.168.0.0 255.255.0.0 192.168.100.2

end
copy running-config startup-config

5. Verifica e Collaudo dell'Infrastruttura

* **Test HSRP**: Facendo cadere il Core 1, il Core 2 subentra istantaneamente mantenendo attivi i gateway virtuali.

* **Test NAT (`show xlate`)**: Inviando traffico da un qualsiasi PC (Sede 1 o Sede 2) verso l'esterno, l'ASA traduce dinamicamente l'IP sorgente sull'interfaccia outside.